

- Assess if it has an optimal solution, if it is unbounded, or if it is infeasible.
- Write the Lagrangian, the dual function and the dual problem.
- Assess if the dual problem has an optimal solution, if it is unbounded, or if it is infeasible.

$$\min_{x \in \mathbb{R}^2} -4x_1 + 2x_2$$

subject to

$$-x_1 + x_2 = 2,$$

$$x_1 - x_2 = 1,$$

$$x_1 \geq 0,$$

$$x_2 \geq 0.$$